

SSM – SOURCE SPECIFIC MULTICAST
– Receiver subscribes using IGMPv3 , providing both source (s) and group (g) to the first-hop router
– No Rendezvous point (RP) required → PIM-SSM builds only (s, g) Short-est Path Trees (SPT)
– No shared tree (*, g) mode used; SSM is source-directed
– IANA reserved 232.0.0.0/8 for SSM in IPv4
– Join messages are forwarded hop-by-hop toward the source to establish forwarding path
– Uses unicast routing table (RPF) to maintain loop-free delivery
PIM SPARSE-DENSE MODE
PIM Sparse Mode: Pull model – multicast traffic is forwarded only on request
PIM Dense Mode: Push model – traffic is flooded everywhere, then pruned
Sparse-Dense Mode: Supports both modes per multicast group; choice depends on RP availability

PIM-DN vs. PIM-SM
PIM is protocol-independent: it relies on the unicast routing table for RPF checks and to forward joins toward the source or Rendezvous point (RP).
– PIM Dense Mode (DM)
– “Push” model
– Floods multicast traffic throughout the network
– Prunes back where traffic unwanted
– PIM Sparse Mode (SM)
– “Pull” model
– Traffic sent only on request
– Requires explicit <i>Join</i> messages

Usage Recommendation:
Dense Mode: Best for small or tightly scoped networks where most devices need multicast
Sparse Mode: Preferred for large-scale or distributed environments where multicast receivers are few or spread out

VXLAN
Issues of L2: STP, Max amount of VLANs (4094), Large MAC Address tables
VXLAN (Virtual-Extensible LAN): Tunnels Ethernet (Layer 2) over IP using MAC-in-UDP encapsulation (Port 4789). For flexible and scalable network segmentation.

VNID (VXLAN Network Identifier): 24-bit identifier (up to 16 million segments) that defines the VXLAN broadcast domain.
VTEP (Virtual Tunnel Endpoint): Device (switch, router, or host) responsible for encapsulating/de-encapsulating VXLAN traffic.
NVE (Network Virtual Interface): Logical interface on a VTEP used for VXLAN tunnel operations.

TUNNEL
– VXLAN establishes IP tunnels between VTEPs to extend Layer 2 networks across Layer 3 boundaries.
– VXLAN enables both L2 and L3 VPN functionality in overlay networks.
– VXLAN traffic is encapsulated in UDP (default port: 4789).

FRAME FORMAT
– Ethernet frame → VXLAN Header → UDP → Outer IP Header.
– The VXLAN header contains the 24-bit VNID and flags.
– Outer headers allow Layer 2 frames to traverse IP overlay networks.

VIRTUAL NETWORK IDENTIFIER (VNI)
– 24-bit VXLAN Network Identifier uniquely defines VXLAN segments.
– Replaces traditional VLAN IDs (12-bit), enabling ~16 million logical segments.
– Used by VTEPs to map traffic into corresponding Layer 2 domains.

VXLAN TUNNEL ENDPOINT (VTEP)
– Connects the overlay (VXLAN) and underlay (IP) networks.
Types:
– Software VTEP: Located on hypervisors using virtual switches.
– Hardware VTEP: Located on routers/switches with ASICs for performance.

Interfaces:
– VTEP IP Interface: Connects to the underlay network and handles encapsulation.
– VNI Interface: Virtual interface per segment (like SVI); handles segregation of Layer 2 domains.

MAC ADDRESS LEARNING
On control plane: happens proactively
On data plane: ad-hoc with flooding

- Each VTEP maintains a VXLAN mapping table linking destination MAC addresses to remote VTEP IPs.
- **Learning via ARP:**
 - Host H1 sends ARP request, switches learn H1’s MAC.
 - ARP request is flooded to H2.
 - H2 responds; switches learn H2’s MAC.
- **Learning Methods:**
 - **Static VXLAN:** Manual MAC-to-VTEP mappings. Doesn’t scale well; BUM traffic is inefficient.
 - **Multicast VXLAN:** VTEPs join multicast groups per VNI. Scales better, offloads BUM replication. 20+ VTEPs = there is too much traffic, doesn’t scale well.
 - **MP-BGP EVPN:** Modern solution using BGP as control plane. Dynamically learns MAC/IP info.

EVPN
Overcome flood-and-learn limitations, doesn’t rely on data plane learning, utilizes router control plane MP-BGP, works with different encapsulation techniques (VXLAN, MPLS), excellent scalability, I2 and I3 Support.

MP-BGP EVPN (MULTIPROTOCOL BGP FOR ETHERNET VPN)
– Enables protocol-based VTEP discovery and host reachability via control plane learning
– Reduces flooding by replacing data-plane learning
– Extends BGP with multiprotocol capabilities (AFI/SAFI)
– Uses MP_REACH_NLRI and MP_UNREACH_NLRI for route advertisement and withdrawal

EVPN ROUTE TYPES
Type 2 – Host Advertisement: Advertises host MAC (mandatory), optionally IP, along with L2VNI and optionally L3VNI. Used for MAC learning, ARP suppression, and host mobility. Sent when hosts connects to VTEP.
Type 5 – Subnet Advertisement: Advertises IP prefix + prefix length with L3VNI. Used for inter-subnet routing. VTEP redistributes connected/static/dynamic IP routes. Additional attributes: L3VNI, extended communities.

HOST DELETION & MOVE
Host Deletion: When a host detaches, its ARP (default: 1500s) and MAC entry (default: 1800s) time out on the VTEP. Upon aging, the VTEP withdraws the host’s MAC/L2VNI and IP/L3VNI advertisements.

Host Move: When a host moves to a new VTEP, the new VTEP advertises updated reachability with a higher move sequence number. The old VTEP withdraws its entry, completing the migration.

ROUTE DISTINGUISHER (RD) VS. ROUTE TARGET (RT)
Route Distinguisher (RD): Uniquely identifies VPN routes → allows same IP prefix to be used in different VPNs. Can be IPv4 or ASN <i>Used to make routes unique in BGP (VPNv4/v6)</i> . Forms VPNv4 NLRI: RD:IPv4 prefix Next Hop Target (RT): Controls route import/export between VRFs. <i>Used as extended BGP community.</i>

- **How RTs Work:**
 - A route is tagged with an RT when advertised by BGP.
 - Other VRFs import the route if the RT matches their import policy.
 - Allows overlapping or shared connectivity between tenants (e.g., shared services).
- **Format:** Typically in the form **ASN:nn** or **IP-nn**, e.g., 65090:100, 1:10
- **Multiple RTs can be used:** A route can have multiple RTs for flexible policies (e.g., one RT for VPN, another for shared services)

EVPN (ETHERNET VPN) – L2
KEY FEATURES
– L2 bridging across L3 networks

BGP Control Plane: Distributes MAC info (no flooding)
VXLAN Overlay: Encapsulates L2 in L3 UDP (data plane)
Multi-Tenancy: via VNI segmentation
Redundancy: All-active multihoming, ECMP, fast convergence

USE CASES
– Multi-tenant datacenter interconnects (DCI)
– Extending L2 over WAN between remote sites
– Scalable, segmented L2 fabrics
BGP CONTROL PLANE
– PE’s learn MACs from local CEs (data plane)
– MACs advertised via BGP (control plane)
– Uses Route Distinguishers and MPLS labels
– Remote PE’s update L2 RIB/FIB with MAC and next-hop info
– Enables seamless L2 across IP/MPLS backbone

EVPN NLRI
– EVPN uses MP-BGP with specific AFI/SAFI
– Supports multiple route types and attributes
– Unsupported routes are dropped by BGP

AUTODISCOVERY VIA ROUTE REFLECTORS
– Route Reflector (RR) avoids full-mesh iBGP
– RR reflects iBGP routes to other PE’s
– RR doesn’t participate in EVPN or pseudowires
– RR needs only address-family l2vpn evpn
– L2VPN RIB stores endpoint/VFI info for control plane
– BGP UPDATE from spines contain ORIGINATOR_ID (origin leaf)

HOST DETECTION
– Host connects to VTEP → MAC learned locally
– VTEP advertises MAC as L2VNI via BGP EVPN
– MAC learning follows normal Ethernet semantics

INGRESS REPLICATION (IR)
– BUM traffic, when Multicast underlay network is not used, handle multi-destination traffic (ARP → unicast)

EARLY ARP TERMINATION (ARP SUPPRESSION)
– Avoids flooding ARP requests
– VTEP queries control plane for MAC/IP/VNI mapping
– If known → direct unicast (no broadcast)

SILENT HOST FLOW (FALLBACK)
– If IP/MAC unknown → ARP sent via ingress replication
– Replicated ARP request goes to remote VTEPs
– Only correct host responds → update reflected to all VTEPs
– Future traffic uses updated BGP mapping

VRF – VIRTUAL ROUTING AND FORWARDING
– Multiple isolated routing tables on one device
– Each tenant = one VRF → traffic isolation
– Supports independent policies per tenant
– Key for scaling and multi-customer separation

IRB – INTEGRATED ROUTING AND BRIDGING
– Enables inter-VLAN routing inside EVPN
– Avoids central gateway → no “traffic tromboning”
– Two modes: Symmetric and Asymmetric

SYMMETRIC IRB (L2 + L3)
– Routing/bridging on ingress + egress VTEPs
– Uses L3 Transit VNI (same in both directions), One L3 VNI per VRF (Tenant)
– Scales well; clean separation of MAC and IP

ASYNCHRIC IRB (L2)
– Routing only on ingress, bridging on egress
– VXLAN uses destination VNI in both directions
– One L2 VNI per VLAN/Subnet
– Simple config, but requires all VLANs/VNIs on all VTEPs

DISTRIBUTED ANYCAST GATEWAY (DAG)
– Same gateway IP+MAC on all VTEPs
– Enables local default gateway for hosts
– Supports mobility + optimal forwarding

L3 HOST DETECTION
– Host sends ARP/ND to local VTEP
– VTEP learns MAC/L2VNI and IP/L3VNI
– Info is advertised in EVPN (control plane)

MPLS
– Label Switched Path (LSP) → pre-determined path across MPLS network
– advantage eBGP between PE-CE: No mutual redistribution, same routing process
– encrypt traffic flowing over MPLS L3VPN backbone? yes (e.g. bank)
– Unicast Reverse Path Forwarding (uRPF): checks source of each packet & verifies that source is in routing table
– control plane (e.g. OSPF) → to learn labels
– iBGP used to exchange NLRI (RD, RT, IPv4 Prefix, NextHop &VPN Label) between PE
– mp-null = networks are directly connected, no more label switching

WAN
Connects remote LANs via SPs for data/voice/video; key needs: bandwidth, control, design, resilience, mgmt. **Requirements:**
Bandwidth: App needs, peak usage, reserve for VoIP
Control/Security: Trust provider? No full control
Availability: Redundancy, SLA for failures
Mgmt: Inband vs out-of-band

PRIVATE WAN
Point-to-Point: Leased L2 line (Ethernet); monthly fee; private circuit
Dark Fiber: Physical fiber lease; costly; ISPs prefer selling lambdas
Connection-oriented: Predefined path, packets carry IDs (ATM, Frame Relay)
Connectionless: No setup; full address in each packet (Ethernet, MPLS VPN)

TERMINOLOGY
CE – Customer Edge: no knowledge of MPLS, no labels; connected to PE
PE – Provider Edge: connected to CE; runs iBGP and LDP; uses VRFs
P – Provider or LSR(Label Switch Router): inside MPLS VPN, no CE connection; forwards labels

DATABASES, PLANES
RIB (Routing IB (Information Base)): Learned prefixes from routing protocols

FIB (Forwarding IB): Built from RIB; only best routes for forwarding
LIB (Label IB): All label mappings; 1 label per prefix
LFIB (Label Forwarding IB): Built from LIB; used for actual forwarding decisions. (LFIB only contains currently best LSP (decision: Routing Protocol))

Control Plane: Builds routing/label tables (RIB, LIB)
Data Plane: Forwards packets (FIB, LFIB); pushes/swaps/pops labels (see below)

MPLS HEADER
4-byte header before IP:
– Label (20b) – actual MPLS label
– EXP (3b) – QoS/CoS, Now called Traffic Class (TC)
– S-bit (1b) – bottom of label stack indicator, 1 = True = last label before IP header
– TTL (8b) – time-to-live (eq total IP TTL)

TTL MPLS
– ingress PE router decrements IP TTL field & copies packet’s IP TTL field into new MPLS TTL
– P routers decrements MPLS TTL
– egress PE router decrements MPLS TTL, pops final MPLS header, copies IP TTL
– traceroute receive ICMP Time Exceeded, Provider doesn’t want to expose MPLS network to fix: disable MPLS TTL propagation (on PE), PE set MPLS TTL = 255, egress leaves PE original IP TTL unchanged
– = MPLS network appears as single router hop from TTL perspective

LABEL DISTRIBUTION PROTOCOL (LDP) – CONTROL PLANE
– Distributes labels to neighbors using control plane
– Hello messages: Sent via UDP (Port 646) to 224.0.0.2 to discover neighbors
– TCP (Port 646) connection is used to exchange label bindings (prefix to local label)
– Routers advertise all local bindings after TCP session is up
– Label mapping used to build LIB → LFIB
– LDP router ID must be reachable (via routing table)
– Each router manages local labels independently

MPLS L3 DATA PLANE
– VPN traffic uses 2 labels (stacked):

Outer label: Transport label (LDP); identifies LSP between ingress/egress PE
Inner label: VPN label (MP-BGP); identifies customer VRF
Push: Ingress PE; classify and label packets
Swap: P router; replaces label, forwards based on new label
Pop: Egress PE removes label; sends original packet to CE
Penultimate Hop Popping: MPLS feature, penultimate router removes the outer MPLS label before forwarding to egress PE, default enabled, ISPs disable it

VRF TABLES
– VRF = Virtual Routing and Forwarding table (Virtual router inside a PE. Maintains isolated RIB + FIB per customer)
– Stores separate routing info per customer (VPN isolation)
– Exists per MPLS-aware PE router; one per attached customer
– Contains: RIB, FIB, and separate routing process per CE

VPNv4
– 64-bit RD + 32-bit IPv4 = 96-bit VPNv4 prefix
– transferring VPNv4 between PE router → Multiprotocol iBGP (MP-iBGP)

OVERLAY TECHNOLOGIES
MODERN PROVIDER NETWORK
MPLS: Label-based forwarding (fast, scalable)
LDP: Distributes labels for MPLS paths
IGP: Underlay routing (e.g., OSPF, IS-IS)
MP-BGP: Extends BGP to carry VPNv4/v6, EVPN routes

DRAWBACKS OF TRADITIONAL NETWORKS
Control Plane: LDP/RSPV-TE adds complexity
Scalability: Per-flow/path state limits growth; LSP and signaling overhead increase rapidly
OAM:

- **Troubleshooting:** Traceroute less useful in MPLS; labels hide topology
- **Traffic Eng.:** LDP lacks TE; relies only on IGP cost

Fast Reroute: Limited coverage; microloops possible

SEGMENT ROUTING (SR)
Source routing: Sender defines full path using Segment List (Segment = Instruction)
State = Segment Identifier: Each SID = 1 instruction (e.g., forward via ECMP, specify interface, or to a service)
State in packet: No per-flow state in network; intermediate nodes follow SID instructions
No new control plane: Uses existing protocols (OSPF, IS-IS, BGP) with extensions; no LDP or RSVP-TE needed
Segment List: Ordered SID list carried in packet header; defines full route
Simple but powerful: Enables TE, fast reroute, policy routing

SEGMENT LIST OPERATIONS
Push: Insert SIDs into packet; set active SID (top of list)
Continue: Active SID not yet completed; keep processing it
Next: Current SID completed; activate next SID in list

SEGMENT SIGNIFICANCE
Global Segments: Known and supported by all SR nodes in the domain, Installed in forwarding tables across the network (e.g. “Forward packet according to shortest path to Node1”)
Local Segments: Defined and installed only on originating node, Not forwarded by others, but must be understood network-wide (e.g. “Forward packet on interface to Node2”)
Global segments: Defined in the SR Global Block (SRGB) and should be consistent across all nodes; **local segments** are defined in the SR Local Block (SRLB) and are specific to the local SR node.

SR CONTROL PLANE SEGMENT TYPES
IGP Prefix Segment: Global SID tied to IGP prefix (multi-hop); all nodes install forwarding entries
IGP Node Segment: Global SID for a specific node (shortest-path forwarding)
IGP Anycast Segment: Global SID for a group of nodes; traffic sent to nearest IGP Anycast Segment
L2 Adjacency SID: Local SID for Layer-2 segment (e.g., Ethernet link)
Combining Segments: – End-to-end paths can mix IGP and BGP segments – Traffic to BGP Anycast → more ECMP in data centers
SR MPLS
– Reuses existing MPLS data plane → no hardware change needed
– Segments = MPLS labels; Segment List = label stack (top = active)
– Segments distributed via IGP/BGP; no LDP required (interoperable if needed)
– Supports both IPv4 and IPv6 networks
BENEFITS OF SEGMENT ROUTING
Benefits: Simplification (removes protocols, simple operations, admin and mgmt), enhanced Traffic eng. (Delay, Bandwidth, Packet Loss, TE metric, Controller, Source-Node), Seamless deployment, Robust, Network Innovation (B Container Networking)
Source Routing: Balances distributed intelligence with centralized optimization
TL-FRA: Fast reroute technique; protects against link/node failure with microloop avoidance and no pre-calculation dependency
Traffic Engineering (TE): Optimizes network performance by analyzing and controlling data flow to reduce congestion and improve QoS
Service Function Chaining (SFC): Chains SDN services in order; automates traffic between VNFs and optimizes routing for performance

QoS
Internet is **best effort**: no guarantees, no QoS; all traffic treated equally (net neutrality); simple, scalable, but no delivery/order assurance or prioritization

QOE & ROUTE PINNING
QOE – Quality of Experience: Perceived service quality from user perspective

Route Pinning: Keeps flow on a fixed path to prevent oscillation (don’t switch immediately to “better” path)

NETWORK PERFORMANCE METRICS
Latency / Delay [ms]: Time for packets to travel src → dest (Voip < 150ms)
End-to-end Delay: Total time sender to receiver
One-Way Delay: From first bit sent to last bit received
Delay Components: Transmission delay (time to push onto link), Processing delay (lookup, queuing), Propagation delay (physical travel time)
Jitter [ms]: Variation in delay between packets, caused by re-routing/queuing (Voip>30ms, Calc: no queue - queued delay)
Throughput: Rate of successfully delivered data
Packet Loss [%]: Dropped packets due to congestion or errors (Voip < 1%)
Bandwidth [Gbit/s]: Maximum transfer capacity of a link

QUEUEING ALGORITHMS
FIFO (First-In-First-Out): Basic, no prioritization
Priority Queueing (PQ): Multiple queues, serve highest first; others may starve
Round-Robin: One packet per queue in turn (fair, but ignores priority)
Weighted Fair Queueing (WFQ): Round-Robin with weights, e.g., 2 packets from Q1, 4 from Q2
Class-Based WFQ (CBWFQ): WFQ with user-defined classes, queue limits, max bandwidth guaranteed or max % of bandwidth (logical queues based on IP Precedence only)
Low Latency Queueing (LLQ): Adds strict priority queue (priority class) to CBWFQ for delay-sensitive traffic (e.g. voice) (based on IP Precedence, DSCP, src, port, protocol,...)

QUEUE MANAGEMENT
Tail Drop: Drops packets when queue full; huge interruption of traffic → same as no connectivity
TCP Global Sync: Many TCP flows back off and restart simultaneously → link underutilization
TCP Starvation: TCP slows down after drops, UDP doesn’t → queues filled with UDP; TCP squeezed out
RED: Random early drops before full queue to prevent global sync and TCP collapse. Dropped TCP segments cause TCP sessions to reduce their windows sizes
WRED: RED + DSCP/EXP-based drop logic, prioritizes higher-marked traffic
DSCP / EXP: DSCP (6-bit in IP header) marks packets for QoS; used in Diff-Serv for classifying traffic. EXP (3-bit in MPLS label) serves same purpose within MPLS networks; often mapped from DSCP.

POLICING VS. SHAPING
Policing (Inbound mostly): Drops packets that exceed configured rate limits
Shaping (Outbound): Buffers packets to smooth traffic bursts and conform to profile
QOS MODELS
Best Effort: No guarantees, all traffic treated equally (follows Internet neutral-ity)

Integrated Services (IntServ): End-to-end QoS, per-flow resource reservation, precise but not scalable (uses RSVP)
Differentiated Services (DiffServ): Class-based, scalable approach using marking (e.g.,), no hard guarantees

TRAFFIC MARKING
L3 Marking: ToS byte → DSCP (6 bits) + IP Precedence (3 bits)
L2 Marking: Dot1q label → 802.1p CoS bits
MODULAR QOS CLT (MQC)
Class Map: Define traffic classes (e.g., match voice or video)
Policy Map: Define actions for each class (e.g., limit, shape, priority)
Service Policy: Apply policies to interfaces or directions (in/out)

CDN
Origin Server: Central content source (original files), usually in a datacenter
Edge / CDN Server (POP - Point of Presence): Geographically distributed, caches content

DNS Infrastructure: Directs users to optimal edge server (e.g. via Geo-Routing)
KEY BENEFITS
Latency Reduction: Nearby edge servers reduce round-trip time
Availability: Failover and redundancy in case of node failure
Scalability: Handles traffic spikes via load balancing
Cost Optimization: Reduces backend and transit load on origin
DDoS Protection: Edge servers absorb attacks → not all traffic on one server
Global Load Reduction: Less long-distance traffic across the Internet

REQUEST ROUTING TECHNIQUES
– Decides which edge server should serve a client request
– Goal: Best performance (e.g. proximity, load, responsiveness)
DNS-BASED GEO-ROUTING
– Each edge has a unique IP
– DNS server picks closest/optimal edge server based on: – Resolver IP location (not user!) – GeoIP DBs (MaxMind, IP2Location), load, latency, business rules
– Limitation: DNS Resolver != user location → can cause wrong choice
EDNS(0) AND CLIENT SUBNET EXTENSION (ECS)
– (Resolver includes part of client IP in DNS request (e.g. /24 subnet)
– Authoritative DNS makes better decision based on actual client region
– Improves accuracy without revealing full IP
ANYCAST WITH BGP
– Same IP (e.g. 7.7.7.7) advertised from multiple locations
– BGP routing decides which path is “best” (AS-path, local pref, etc.)
– No DNS logic or per-client decision → pure BGP convergence

Pros: Fast failover, simple, no app logic needed
Cons: Less control, BGP is best latency, route flapping risk

HTTP CACHING & HEADERS
– Caching is controlled via HTTP headers between clients, proxies, and servers

Cache-Control: Main directive (**no-cache, no-store, max-age, must-reval-idate, etc.**)
Expires: Absolute expiration time (older method, replaced by **Cache-Control**)
Etag: Validator tag (version/hash), used with **If-None-Match**
Last-Modified: Timestamp used with **If-Modified-Since** for revalidation
Age: Time (in seconds) since response was fetched from origin
Validation: Client uses **Etag** or **Last-Modified**; server returns 304 if unchanged

OTHER INFOS
AD: Inter-protocol choice (e.g., OSPF vs RIP) → lower wins
Cost/Metric: Intra-protocol choice (e.g., OSPF Static Interface) 1
Static (Next Hop): 1
BGP External: 20
EIGRP Internal: 90
OSPF: 110
ISIS: 115
RIP v2/v2: 120
EIGRP External: 170
BGP Internal: 200

EVPN BGP ROUTING TABLE INFOS
* = Would not be there if it was L2 VNI BGP Routing Table
Route Distinguisher: 172.16.255.101:32777
Route Type: 2
MAC Address Length: 48
MAC Address: 5254.00f8.29a8
IP Address Length: 32
IP Address: 10.10.0.100
L2 VNI: 30010
L3 VNI: 50000
Remote VTEP IP Address: 172.16.254.101
L2 VTEP Target: 1:10
L3 Route Target: 65000:50000

Leaf-03# show bgp l2vpn evpn 10.10.0.100
BGP routing table information for VRF default, address family L2VPN
Route Type: 2
Route Distinguisher: 172.16.255.101:32777
BGP routing table entry for
[2]:[0]:[0]:[0]:[48]:[5254.00f8.29a8]:[32]:[10.10.0.100]/272, version 19897
Paths: (1 available, best #1)
Flags: (0x000202) (high32 00000000) on xmit-list, is not in L2vri/evpn, is not in HW
Advertised path-id: 1
Path type: internal, path is valid, is best path, no labeled nexthop
Imported to 2 destination(s)
AS-Path: NONE, path sourced internal to AS
172.16.254.101 (metric 85) from 172.16.255.1 (172.16.255.1)
Origin IGP, MED not set, localPref 100, weight 0
Received label 30010 50000
ECMcommunity: RT:1:10 RT:45000:50000 ENCAP:8 Route Map:5254.00ca.69ae
Originator: 172.16.255.101 Cluster List: 172.16.255.1

TODO:
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